PakEmis Sector Technical Note TN01

A Reproducible Agricultural Greenhouse-Gas and Air-Pollutant Inventory for Pakistan with Mass-Conserving Spatial Allocation

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Abstract

This technical note documents the agricultural contribution to PakEmis using a common livestock, crop, fertiliser and nitrogen-flow account for greenhouse gases and air pollutants. The supported national core covers July 2015–June 2025. FY2023–24 remains the detailed reference year; earlier long calendar-year diagnostics are separate products and do not define the common fiscal period.

The reference TAN-coherent configuration produces 223 Mt CO_2e on AR5 GWP-100, 2619 kt NH_3, 318 kt NO_2-equivalent NO_x and 54.13 kt primary $PM_{2.5}$ in FY2023–24. FY2024–25 uses observed fiscal crop and nutrient controls and the consistent census-anchored livestock reconstruction. Native gas masses, pollutant masses and GWP aggregates remain separate.

Species- and stage-resolved TAN losses drive both NH_3 and the surviving manure pools used for N_2O. An IPCC volatilisation configuration is retained separately; it is not mixed probabilistically with the TAN reference. Conditional uncertainty replays shared herd, crop, nitrogen and applicable fate-factor draws, while management shares, factor applicability and missing sources remain distinct limitations. Direct comparison with the publicly released HTAPv3.2 monthly country tables matches exact July–June periods and records shared REAS/EDGAR inputs. The release provides national activity detail and process-matched geographic support beyond a simple clipped global inventory.

Keywords: agricultural greenhouse-gas inventory, air-pollutant emissions, ammonia, hybrid IPCC Tier 1/2, dasymetric mapping, gridded livestock, mass-balance reconciliation, district downscaling, Monte-Carlo uncertainty, Pakistan, MRV, transparency reporting

PakEmis Sector Technical Note TN01. Technical documentation of one sector module of the PakEmis release identified below, deposited with the data as part of the documentation record. The national data-description article (`paper_national/`) is the only document that states the national totals as its own result.

Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

Release version	`1.0.0rc69`
Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
Source manifest	`c1be6d314cfce1ea` (SHA-256 of `data/manifests/source_manifest.csv`, first 16)
Data-description article	`paper_national/` (ESSD, in preparation), same release

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1. Introduction

National greenhouse-gas inventories provide the accounting basis for reporting under the Paris Agreement. Under the Enhanced Transparency Framework, countries regularly report sectoral emissions using Common Reporting Tables. In many developing countries, agriculture, forestry and other land use (AFOLU) accounts for a substantial share of national emissions but is also among the sectors with the weakest activity data.

Pakistan illustrates this problem clearly. Agriculture is one of the country's largest emitting sectors, yet its national inventory provides limited spatial detail. Some livestock estimates also continue to rely partly on projections from older censuses. National totals are necessary for international reporting, but they provide little information about where emissions occur or where mitigation and improved data collection should be targeted.

Agricultural activity also produces pollutants that affect air quality. Ammonia released from fertiliser and manure is the principal alkaline precursor of secondary inorganic particulate matter. Agricultural NH_3 can therefore contribute to $PM_{2.5}$ concentrations far from the fields and livestock systems where it originates. Crop-residue burning in the Punjab rice–wheat belt also releases primary $PM_{2.5}$, carbon monoxide and ozone precursors during the post-monsoon period, when particulate pol-

lution in Lahore is particularly severe.

Despite this importance, Pakistan has no gridded agricultural air-pollutant inventory derived from national activity data. Air-quality modelling instead relies largely on global emissions products. These products provide valuable regional coverage, but their activity data and spatial assumptions cannot reliably distinguish, for example, an intensively cultivated canal-belt district from an extensive rangeland district.

A useful national system must therefore meet three requirements. First, it must be reproducible and updateable as new official statistics become available. Second, it must provide provincial, district and gridded estimates that support subnational planning and air-quality modelling. Third, its greenhouse-gas and air-pollutant calculations must use consistent activity data and nitrogen flows.

The third requirement is physically important. A unit of manure nitrogen may be lost as ammonia during housing, storage, grazing or field application, or it may remain in the system and contribute to direct or indirect nitrous-oxide emissions. Calculating these pathways in separate workbooks creates a risk of double counting, omission, or inconsistent system boundaries. Similar inconsistencies arise when a national inventory, a spatial-allocation exercise, and an emissions input for a chemistry model are developed independently. The resulting products may appear detailed while failing to reproduce the same national total or physical flow.

The same argument applies one level up. An agricultural inventory that is internally consistent may still be inconsistent with the sectors around it, because several quantities are genuinely shared: nitrogen leaving the manure-management balance as household fuel has to arrive somewhere, and crop-residue burning has to be distinguishable from other open burning. The framework described here is the agriculture module of a ten-sector national inventory (Ahmad, 2026), in which agriculture is 50.2 % of the national total and those transfers are executable assertions rather than statements in a methods document. This paper documents the sector; the companion documents what holds between sectors, and neither repeats the other.

Many of the required inputs already exist. Pakistan publishes national and subnational agricultural statistics, while open global datasets provide spatial information on livestock, crops, land cover and fire occurrence. The remaining need is an integrated framework that calculates emissions, allocates them spatially under an explicit mass-balance constraint, evaluates the resulting products, and propagates uncertainty from a common set of parameters.

This study develops such a framework for Pakistan. We construct a *hybrid Tier 1/Tier 2* agricultural inventory using the 2006 IPCC Guidelines and the 2019 Refinement. We then allocate individual emission streams to provinces, a 0.05° grid, and administrative districts using process-specific spatial proxies. Uncertainty is quantified using analytical propagation and Monte-Carlo simulation.

The central design principle is that the calculated inventory total serves as the control value. Every provincial, gridded and district product is constrained to reconcile with that control. Emission factors and activity data therefore determine the quantity emitted, while spatial proxies determine only where those emissions are allocated.

1.1. Contributions

This paper makes five main contributions.

First, it provides a reproducible national agricultural inventory implemented in a single codebase that runs on the Python standard library alone. Emission factors, activity values and other parameters are stored in versioned data files rather than embedded in the calculation code. The inventory covers six IPCC categories: enteric fermentation (3.A), manure management (3.B), rice cultivation (3.C), managed soils (3.D), field burning of agricultural residues (3.F), and urea application (3.H). Liming emissions under category 3.G are not estimated: no national lime-use activity figure has been sought. All six categories use Tier 1 equations with country-specific activity data. Selected livestock parameters, including animal mass and manure-management-system shares, provide partial Tier 2 characterisation. Tier status is therefore reported separately for each category.

Second, the framework allocates emissions to provinces, a national grid and administrative districts under an explicit reconciliation constraint. Within each control area, every emission stream is distributed in proportion to a process-matched proxy and then rescaled so that the allocated values sum exactly to the control total. Livestock emissions are allocated using gridded livestock density and, where available at district level, census head-counts. Rice emissions are allocated using mapped paddy area, while residue-burning emissions are allocated using satellite fire detections.

Third, the inventory is evaluated at several scales. National category totals are compared with FAOSTAT, EDGAR and Climate TRACE using matched years and category definitions. District patterns are evaluated against independent government crop-area records. The rice allocation has a Spearman rank correlation of 0.92 with official district rice area, while the managed-soils allocation reaches 0.84 against district wheat area. Parametric uncertainty is estimated using both IPCC Approach 1 propagation and Monte-Carlo aggregation. These methods apply different propagation formulations to the same underlying uncertainty assumptions.

Fourth, the framework produces a national reconstruction for 1990–2024 and a district-level historical panel based on fixed spatial shares. The national series reflects changes in activity over time, while the district panel distributes each year's national total according to the audited 2024 district pattern.

These are two different products on two different bases and are not interchangeable. The canonical panel is the ten fiscal years FY2015–16 to FY2024–25, computed on Pakistani fiscal activity with the reference year anchored to the 2024 census; it is what the release reports, reconciles and prices uncertainty for. The 1990–2024 series is a contextual reconstruction on FAOSTAT calendar-year activity, provided for long-run trend and for the Common Reporting Table, and it is not the inventory's reporting basis. Matched-year differences separate the adopted

stock/activity reconstruction from the additional cotton-residue scope. Years lacking an archived calendar comparator remain explicitly unpaired; no absent comparator is replaced by zero.

Fifth, the framework calculates agricultural air-pollutant co-emissions, including NH_3, NO_x and the pollutants released by crop-residue burning. These estimates use the same activity data and process-specific spatial proxies as the greenhouse-gas inventory. A shared, stage-resolved nitrogen ledger aligns the system boundaries used by the IPCC and EMEP/EEA methods and helps diagnose differences between the resulting estimates. The ammonia inventory is also compared with an independent gridded inventory for South Asia. An exploratory Tier-1 land-sector extension is presented separately in Supplementary Note S1 and is excluded from all agricultural results.

Figure 1 summarises the framework. Section 2 reviews relevant inventory and spatial-allocation methods. Section 3 describes the study area and data. Sections 4–6 present the greenhouse-gas inventory, spatial allocation, uncertainty analysis and shared nitrogen accounting. Section 7 describes the evaluation framework. Sections 8 and 9 report the greenhouse-gas and air-pollutant results. Sections 10–12 discuss the findings, limitations and conclusions.

2. Related Work

2.1. National inventory methods

National agricultural inventories generally follow the 2006 IPCC Guidelines (IPCC, 2006) and, increasingly, the 2019 Refinement (IPCC, 2019), which updated several agricultural emission factors and introduced climate-specific disaggregation for soil nitrous oxide.

The IPCC framework defines several methodological tiers. Tier 1 combines default emission factors with national activity data. Tier 2 replaces selected defaults with country-specific parameters or more detailed activity characterisation. Tier 3 uses higher-order models or direct national measurement systems.

Pakistan's national agricultural inventories have been prepared mainly with Tier 1 methods and national activity totals (Global Change Impact Studies Centre, 2019; Mir et al., 2021). This approach supports comprehensive national reporting but provides limited information about subnational patterns or individual processes.

2.2. Global gridded inventories

Several international datasets provide independent estimates of agricultural emissions for Pakistan. FAOSTAT calculates national agricultural and land-use emissions from FAO activity data using mainly IPCC Tier 1 methods and reports results by category and gas (FAO, 2025; Tubiello et al., 2013).

EDGAR provides national totals and gridded emissions at approximately 0.1° resolution using a globally consistent methodology (Crippa et al., 2025, 2020); this paper's comparison table keeps one row from the prior 2024-vintage booklet (Crippa et al., 2024) because the 2025 release revised Pakistan's agricultural CH_4 history and the revision itself is informative. Climate TRACE reports country- and subsector-level agricultural emissions using a combination of inventories, models and observation-informed methods (Climate TRACE, 2026).

These products provide useful external reference points, but they are not substitutes for a nationally compiled inventory. They generally use globally harmonised or model-derived assumptions and may not incorporate the most appropriate national activity data or system definitions. In this study, they are used as external benchmarks rather than as reported national values.

2.3. Spatial allocation of emissions

Allocating national or subnational emissions to a grid is a dasymetric-mapping problem. An emission total is distributed according to a spatial proxy that represents the location or intensity of the underlying activity. Similar approaches are widely used in fossil-CO_2 inventories and global emissions gridmaps (Oda et al., 2018; Crippa et al., 2020).

Several open datasets provide relevant agricultural proxies. The Gridded Livestock of the World (GLW 4) dataset provides modelled species-specific livestock densities (Gilbert et al., 2024, 2018). The Spatial Production Allocation Model (MapSPAM) provides crop-specific physical areas (International Food Policy Research Institute, 2024). ESA WorldCover provides a 10 m land-cover classification (Zanaga et al., 2022), while NASA FIRMS and other satellite products identify active fires and burned areas (NASA, 2025; Giglio et al., 2016).

The quality of a spatial inventory depends on whether each proxy represents the process being allocated. Livestock density is appropriate for emissions produced directly by animals, paddy area for rice methane, cropland for fertiliser-related emissions, and satellite fire occurrence for the location of residue burning. A single proxy applied to an entire inventory category may misallocate emissions when that category contains several distinct activity streams.

A further requirement is mass conservation. Some spatial inventories distribute emissions without subsequently rescaling the mapped values to the original control total. In such cases, the gridded product may no longer reproduce the national or provincial inventory. The framework developed here treats reconciliation as an invariant: every mapped emission stream must sum to its corresponding control total.

2.4. Agricultural ammonia inventories

Agricultural ammonia inventories have largely developed within the air-quality community, separately from greenhouse-gas inventories used for climate reporting. The EMEP/EEA Guidebook represents livestock ammonia through a stage-resolved mass flow. Total ammoniacal nitrogen is tracked through grazing, housing, yards, storage and field application, with emission factors varying by animal category and manure-management system (EMEP/EEA, 2023a).

This process resolution is needed because ammonia losses occur at specific stages of manure handling. By contrast, Tier 1

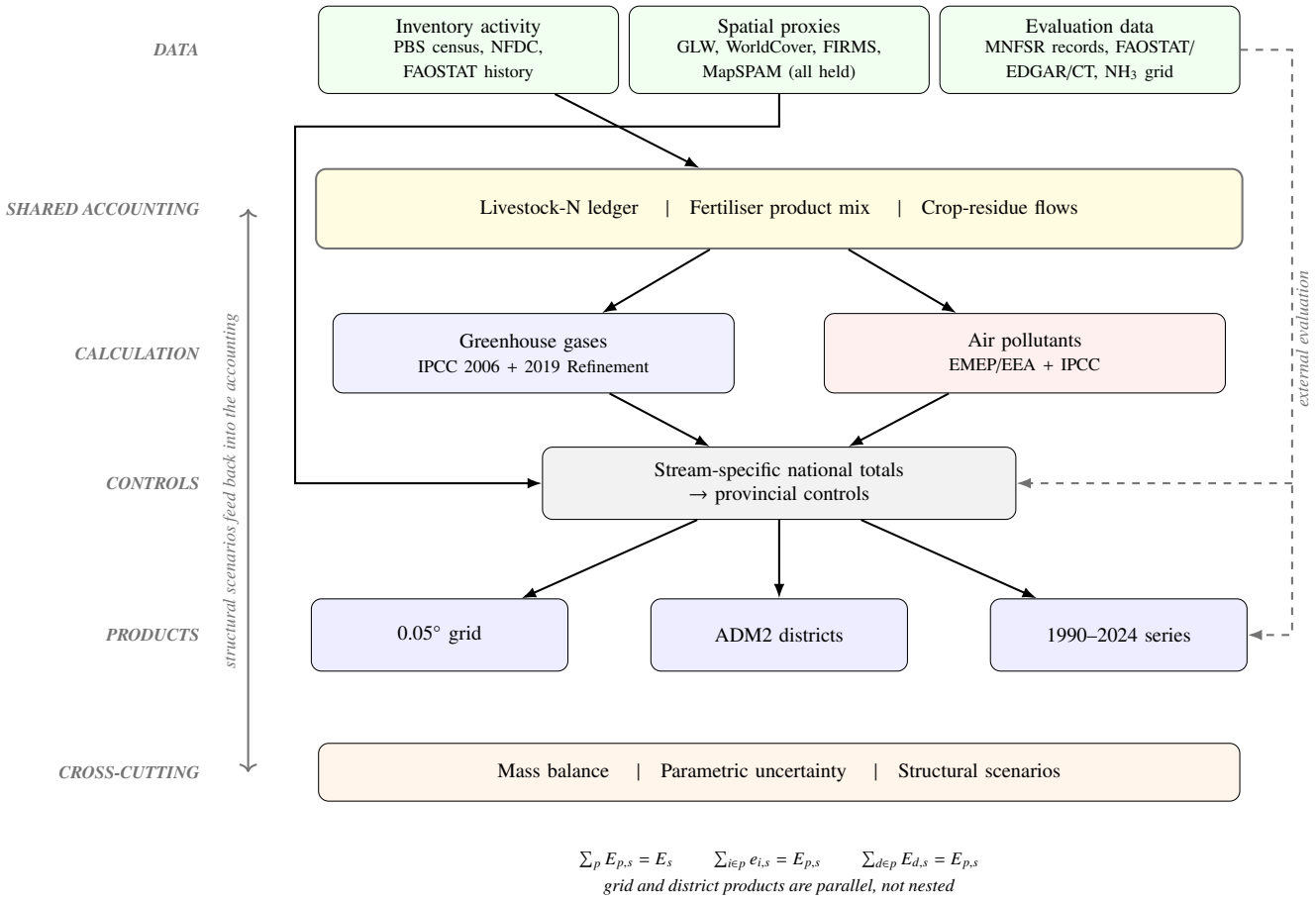


Figure 1: Official activity data feed a shared livestock-nitrogen, fertiliser-nitrogen and crop-residue accounting system. Separate greenhouse-gas (CH_4, N_2O, CO_2) and air-pollutant (NH_3, NO_x, $PM_{2.5}$, CO, NMVOC, BC, SO_2) calculations produce stream-specific national totals, which become provincial controls and are allocated independently to a 0.05° grid and to ADM2 districts on process-matched proxies; every product reconciles to its control. Dashed arrows denote external evaluation, which is applied to the products rather than to the calculations. Evaluation data are never used as inventory inputs.

greenhouse-gas methods often represent manure nitrogen using annual fractions that aggregate across several stages. Combining the two frameworks therefore requires careful alignment of their system boundaries and nitrogen pools.

Tomlinson et al. (2026) provide a regional reference for South Asia. Their framework estimates agricultural NH_3 emissions for the SAARC countries across five sectors using an explicit nitrogen-flow model and provides an openly available 0.1° grid (Tomlinson et al., 2025). We use this inventory to compare both the national magnitude and the spatial pattern of our ammonia estimates.

Other global products, including EDGAR and HTAP, also cover Pakistan. However, they are built without national activity data and rely partly on the same underlying FAO statistics used by many national and regional inventories. They therefore provide useful bounds and methodological comparisons, but agreement with them does not constitute fully independent confirmation.

2.5. Crop-residue burning inventories

Crop-residue burning can be estimated using two broad approaches. Bottom-up inventories calculate the mass burned from crop production, residue-to-product ratios, dry-matter fractions, combustion factors, and the proportion of residues burned. National greenhouse-gas inventories generally use this approach, and the burned fraction is its dominant uncertainty.

Top-down or observation-based approaches use satellite fire detections, fire radiative power, or burned-area products. These datasets provide much stronger information about the location and timing of fires, but conversion to burned mass depends on additional assumptions and also carries substantial uncertainty.

This study uses the two types of information for separate purposes. The national quantity burned is calculated using a bottom-up inventory, while satellite fire detections are used only to allocate that quantity spatially. Satellite detections therefore determine the mapped distribution but do not provide an independent estimate of the national total, a distinction that matters because the two are sometimes conflated.

2.6. Contribution relative to previous work

Compared with conventional national inventory workbooks, this framework integrates calculation, uncertainty analysis, spatial allocation, external evaluation and historical reconstruction in a reproducible codebase. Compared with global gridded

products, it uses nationally grounded activity data and province-level control totals within a hybrid Tier 1/Tier 2 agricultural inventory.

Its main contribution is the combination of these elements. The national inventory establishes the emission totals, process-specific proxies determine their spatial distribution, and exact reconciliation keeps the provincial, gridded and district products consistent with the reported figures. To our knowledge, this combination has not previously been implemented for Pakistan's agricultural sector.

3. Study Area and Data

3.1. Study area

Agriculture in Pakistan is concentrated in the irrigated Indus basin, which extends through Punjab and Sindh and supports most of the country's crop and livestock production. Approximately four-fifths of Pakistan's cropland is irrigated through canals and tube wells. The country also has a large livestock population, with buffalo accounting for a substantial share, and is among the world's largest consumers of urea fertiliser.

These characteristics shape both the composition and the geography of agricultural emissions. Enteric fermentation and soil nitrous oxide dominate the greenhouse-gas inventory, while emissions are spatially concentrated in the irrigated canal belt.

3.2. Activity data

National activity data are drawn primarily from official Pakistani sources. Livestock populations are taken from the 2024 PBS 7th Agricultural Census (Pakistan Bureau of Statistics, 2025). The Pakistan Economic Survey (Ministry of Finance, 2024) is used as a cross-check. The two sources are not expected to match exactly because the census is based on enumeration, whereas the Economic Survey includes projections derived from earlier census rounds. Within the census itself the national summary table and the district tables are separate transcriptions, and they differ by a handful of head in two species. Neither is edited to force agreement: the national table is the emissions control, the district tables drive the spatial allocation, and the residual per species is published with the normalisation factor it implies in `outputs/agriculture_livestock_reconciliation_2024.csv`.

Crop area and production use the full-year historical rows of the Pakistan Economic Survey 2025–26 (Finance Division, Government of Pakistan, 2026), with the same source vintage applied across the core. Partial current-year figures are excluded from annual controls. Fertiliser nutrients and products use the applicable full-year National Fertilizer Development Centre reviews, including FY2023–24 and FY2024–25 (National Fertilizer Development Centre, 2024, 2025). The adopted census anchors a consistent livestock reconstruction through the core, with post-census growth identified as extrapolation. The separate 1990–2024 FAOSTAT calendar reconstruction remains a contextual comparison (FAO, 2025).

3.3. Geospatial data

Spatial allocation uses the open global datasets listed in Table 1.

Livestock emissions are allocated using species-specific densities from Gridded Livestock of the World 4 (Gilbert et al., 2024). Rice-related emissions ride MapSPAM 2020 physical rice area (International Food Policy Research Institute, 2024): the raster (V2r2) is held and hash-pinned since the spatial facility program, its former cropland-fallback class deleted from the allow-list so a checkout without it refuses the build, and the own-year GCD-Rice 2023 classification is published beside it as a crosscheck and a priced district-level bracket. Crop-residue burning uses NASA FIRMS VIIRS active-fire detections matched to the reporting period's crop-burning windows (NASA, 2025). The held record covers all ten fiscal years FY2015–16 through FY2024–25, with 1 734 158 unique detections. FY2024–25's windows come from NASA's keyless country-year files for 2024 and 2025, the same VIIRS standard product for Pakistan only, because the area record behind the earlier nine years ends in June 2024.

Temporal and spatial fire allocations use the same attribution ladder: the crop burning window, the Pakistan mask, cropland, and exclusion of detections within 0.01° of registered kilns. Retained detections are weighted by fire radiative power, with no additive offset. The counts excluded at each step remain in the source diagnostic. Each supported year's detections shape its own allocation rather than a pooled climatology; zero-weight or unsupported locations use a declared fallback. The fire proxy controls allocation, not annual burned mass. Other cropland-related streams are allocated using the cropland fraction from ESA WorldCover 2021 (Zanaga et al., 2022).

The primary provincial and district boundaries use geoBoundaries (Runfola et al., 2020); legacy boundary and derived alternatives are separated under their applicable licences. District allocations are evaluated against crop-area statistics published by the Ministry of National Food Security and Research (Ministry of National Food Security and Research, 2023). National emission totals are compared with FAOSTAT, EDGAR and Climate TRACE.

3.4. Status of the estimates

The estimates remain provisional wherever this inventory holds no nationally measured parameter. The livestock population is anchored to the 2024 agricultural census, but the dairy-cattle share is derived from FAOSTAT milk-animal counts. Rice harvested area and several agronomic fractions are based on published literature or default assumptions pending national survey data. Each such parameter is identified in the input files and carried through the resulting outputs and uncertainty analysis.

4. Methods

The framework consists of four main stages: calculation of the national agricultural inventory; allocation of national totals

Dataset	Use	Role
PBS 7th Ag. Census	activity	livestock, crop area
Economic Survey	activity	crops, livestock cross-check
NFDC	activity	synthetic nitrogen
FAOSTAT (QCL/RFN/GT)	activity/bench.	time series, benchmark
EDGAR 2024/2025	benchmark	matched-boundary national comparisons
Climate TRACE v7.1.9	benchmark	matched-boundary national comparisons
MNFSR district crops	records	district validation (rice, wheat)
GLW 4 (2020)	proxy	livestock density
MapSPAM 2020	proxy (held, hash-pinned)	rice area
ESA WorldCover 2021	proxy	cropland fraction
NASA FIRMS (VIIRS)	proxy	fire occurrence
geoBoundaries / GADM	boundary	province / ADM2 district polygons

Table 1: Principal datasets used in the inventory. Pakistani official statistics provide the national and administrative activity data. Open global products provide spatial proxies and external benchmarks. Inputs used only in the exploratory land-sector extension are listed in Supplementary Note S1.

to provinces; allocation to a 0.05° grid and to administrative districts; and uncertainty, key-category, completeness and quality-control analysis.

The inventory is predominantly Tier 1. Each category uses the equations in the 2006 IPCC Guidelines, together with relevant updates from the 2019 Refinement, and applies them to country-specific activity data. Selected livestock parameters provide partial Tier 2 characterisation where nationally appropriate values are available. These include typical animal masses, the dairy-cattle share and manure-management-system shares.

Tier status is reported by category in Table 2. This is more informative than assigning one tier to the entire inventory, because the level of country-specific detail differs among categories.

All activity values, emission factors and parameters are stored in version-controlled data files; values are not embedded directly in the calculation code. Greenhouse gases are converted to carbon-dioxide equivalents using AR5 100-year global-warming potentials — 28 for CH_4 and 265 for N_2O — to match the Common Reporting Table convention. AR6-equivalent totals are reported separately.

4.1. National inventory

Enteric fermentation and manure management. Enteric methane is calculated using the Tier 1 form of IPCC Eq. 10.19,

$$\mathrm{CH_4^{ent}} = \sum_t N_t \, EF_t \times 10^{-6}, \tag{1}$$

where N_t is the population of livestock category t and EF_t its annual per-head methane emission factor.

Methane from manure management is calculated using IPCC Eq. 10.22 with temperature-dependent factors, and direct nitrous oxide from manure management follows Eq. 10.25. Nitrogen excretion is calculated as $N_{\mathrm{rate},t} \times TAM_t \times 365/1000$, where $N_{\mathrm{rate},t}$ is the daily nitrogen-excretion rate and TAM_t is typical animal mass.

The calculation then distributes excreted nitrogen among manure-management systems. Dung burned as household fuel is assigned to the energy sector and excluded from the agricultural inventory to prevent double counting. Nitrogen deposited directly on pasture is assigned to managed soils rather than to manure management.

Category	Method / data	Country-specific elements
Enteric fermentation	T1 method, CS data	species census, dairy split
Manure management	T1/2, CS data	animal mass, Nex, MMS, temp. zone
Managed soils	T1 method, CS data	fertiliser data, dry-climate EFs
Rice cultivation	T1 method, CS data	area, cultivation period
Field burning	T1 method, CS data	crop production, residue params
Urea application	T1 method, CS data	urea offtake; N series before 2024

Table 2: Method-tier status by category. All categories use the Tier-1 (T1) method equations on country-specific (CS) activity data; manure management is partly Tier 2 through country-specific animal mass and management-system shares. No emission factor is Tier 3 (Pakistani field measurement).

Managed soils. Direct soil nitrous oxide follows IPCC Eq. 11.1,

$$\mathrm{N_2O^{direct}} = \left[(F_{\mathrm{SN}} + F_{\mathrm{ON}} + F_{\mathrm{CR}})\, EF_1 + F_{\mathrm{PRP}}\, EF_{3,\mathrm{PRP}}\right] \frac{44}{28}, \tag{2}$$

where F_{SN}, F_{ON}, F_{CR} and F_{PRP} are nitrogen inputs from synthetic fertiliser, organic amendments, crop residues and pasture deposition respectively.

Nitrogen applied to rice is assigned the flooded-rice emission factor $EF_{1,\mathrm{FR}}$. Indirect N_2O from volatilisation and leaching follows IPCC Eqs. 11.9 and 11.10.

For direct N_2O from non-rice cropland, the reference retains the 2019 dry-climate value $EF_1 = 0.005$ as a conditional factor transfer. Aridity alone does not establish its applicability to irrigated cropland. Rice and grazing use their separate source formulations, and aggregate/climate alternatives remain structural sensitivities. Agreement with an external inventory is not used to select the factor.

Rice cultivation, residue burning and urea application. Rice methane is calculated using IPCC Eqs. 5.1 and 5.2,

$$\mathrm{CH_4^{rice}} = \sum_i EF_c \, SF_{w,i} \, SF_{p,i} \, SF_{o,i} \, t_i \, A_i \times 10^{-6}, \tag{3}$$

where EF_c is the baseline daily emission factor, SF_w represents the water regime, SF_p the pre-season water regime, SF_o the effect of organic amendments, t_i the cultivation period and A_i the harvested area.

Emissions from field burning follow the generic biomass-burning equation in the IPCC Guidelines. Urea-application

CO_2 follows Eq. 11.13. The urea mass has two bases, and each year states which it used in `outputs/agriculture_urea_basis.csv`. For the reference year it is the national urea offtake NFDC prints for the fiscal year itself. For every earlier year, which has a total-nitrogen figure and no published product split, it is reconstructed from that series by a constant chemical relation. The reconstruction was formerly applied to the reference year as well, where it fell short of the printed offtake by 0.207 %; a measurement is preferred to an approximation of it wherever one exists, and the approximation is retained, and labelled, where none does.

Liming emissions under category 3.G are not estimated: no national lime-use activity figure has been sought. Liming is recorded as "not estimated" in the completeness assessment.

4.2. Provincial allocation

National emission streams are allocated to the four provinces and a small residual region using activity-specific administrative shares.

Enteric fermentation, manure-management emissions and pasture-deposited nitrogen use provincial livestock-census shares. Rice methane and rice-specific fertiliser nitrogen use provincial rice area. Residue-related emissions use cropland shares. Synthetic-fertiliser N_2O and urea CO_2 use provincial fertiliser-offtake shares.

Applied manure requires a different treatment, because it is generated near livestock but applied to cropland. Its provincial distribution is therefore derived by intersecting cropland with livestock density at the grid-cell level and then aggregating the resulting surface to provinces.

All provincial shares are normalised to sum to one. Provincial estimates therefore reproduce the national control total exactly.

4.3. Gridded allocation and mass-balance reconciliation

Each provincial emission stream is allocated to a 0.05° grid using dasymetric weighting. Let $E_{p,s}$ be the control total for emission stream s in province p, let Ω_p be the set of grid cells within that province, and let $w_{i,s}$ be the spatial-proxy value for cell i. Emissions in each cell are calculated as

$$e_{i,s} \;=\; E_{p,s}\,\frac{w_{i,s}}{\sum_{j\in\Omega_p} w_{j,s}}, \qquad i \in \Omega_p. \tag{4}$$

This construction guarantees that $\sum_{i\in\Omega_p} e_{i,s} = E_{p,s}$. The grid therefore reconciles to the provincial total, and provincial totals reconcile to the national inventory.

Spatial proxies are assigned by emission stream rather than by broad inventory category. This distinction matters because a single category can contain several nitrogen or activity streams with different geographies.

GLW 4 livestock density is used for enteric methane, manure-management emissions and pasture-deposited nitrogen. Species densities are weighted by their per-head emission factors, so that the resulting proxy represents the spatial distribution of emissions rather than unweighted animal numbers.

MapSPAM paddy area is the declared proxy for rice methane and the rice-specific fertiliser stream; the raster (2020 V2r2, hash-pinned) is held and consumed since the spatial facility program, its cropland-fallback class deleted fail-closed, and the own-year GCD-Rice 2023 classification ships beside it as a published crosscheck and a priced district-level bracket (Section 3). FIRMS active-fire detections are used for crop-residue burning. WorldCover cropland fraction is used for synthetic fertiliser, crop-residue inputs to soils and urea application.

Managed soils are constructed by allocating each nitrogen input separately and then summing the resulting stream-level surfaces. This prevents the geography of synthetic fertiliser from being imposed on manure-derived or pasture-derived nitrogen.

Applied manure is allocated using cropland weighted by livestock density. This proxy represents two physical constraints: manure is applied on agricultural land, but transport costs generally keep it close to the herd that produces it.

Where a proxy has no positive values within a province, the corresponding emission stream is distributed uniformly across that province. This fallback preserves the provincial control total.

4.4. District allocation and historical district panel

The same proportional-allocation rule is used to distribute provincial totals among administrative districts. The main output contains 116 districts across the four provinces. Gilgit-Baltistan, Azad Jammu and Kashmir, and Islamabad Capital Territory are represented together as a residual unit, producing 115 reporting units in total.

For most emission streams, district shares are obtained by summing the relevant spatial proxy within each district and normalising those sums within the province.

Livestock categories receive a more direct treatment: livestock emissions are distributed using district head-counts from the 2024 PBS Agricultural Census, every census district resolved through the inventory's canonical district crosswalk (several census districts sum onto one boundary polygon where the boundary file predates a split). The two census units the boundary file does not draw, Cholistan Area and Larkana District, keep their own share as named rows of the district panel and fall to the province surface in the grid; a polygon the census does not reach is a build failure, not a proxy.

The gridded and district products are parallel allocations of the same provincial controls. They are not nested representations of one another. The 0.05° grid uses continuous gridded proxies, whereas the district table substitutes census head-counts for livestock where possible. Both products reconcile to the same provincial and national totals, but summing grid cells within a district will not necessarily reproduce the census-controlled district estimate. Formally, $\sum_{i\in\Omega_p} e_{i,s} = E_{p,s}$ for the grid and $\sum_{d\in p} E_{d,s} = E_{p,s}$ for the district product, and both satisfy $\sum_p E_{p,s} = E_s$.

No consistent multi-year district activity dataset is available. The historical district panel therefore holds the audited

2024 district shares fixed and rescales them to each year's national category total for 1990–2024.

The district values consequently follow the national time series. They do not represent observed changes in the geographic distribution of activity over time. This assumption is evaluated in Section 11.

The exploratory land-sector extension in Supplementary Note S1 is excluded from the agricultural totals, benchmarking, key-category assessment and uncertainty analysis.

4.5. Uncertainty analysis

Uncertainty is estimated using two propagation approaches applied to the same set of activity-data and emission-factor uncertainties.

IPCC Approach 1 propagates uncertainty analytically. Uncertainties in multiplicative terms are combined in quadrature, while uncertainties in additive components are combined using their contributions to the total.

The Monte-Carlo analysis samples the *components*: each engine row is drawn from a mean-preserving lognormal at its own declared width—appropriate for non-negative emissions with potentially right-skewed uncertainty—with one shared deviate per declared correlation group, so rows split apart for allocation reasons cannot average their common error away. The simulation uses 10^4 iterations, and the category and sector totals are the *sums of the sampled rows* in every realisation rather than sampled quantities in their own right.

The two approaches test the sensitivity of the reported interval to the form of propagation. They do not provide independent estimates, because both use the same underlying uncertainty assumptions. Agreement between them therefore indicates consistency between analytical quadrature and lognormal aggregation.

Managed-soils components that share an emission factor are treated as perfectly correlated. This applies to separate volatilisation and leaching streams that appear as distinct rows because they use different spatial proxies. Treating them as independent would artificially reduce uncertainty when one physical stream is divided into several reporting components.

Other within-category and cross-category correlations are not modelled *by this sector-scoped stream*, and the qualifier matters because a second engine treats two of them differently. This sector's Monte Carlo samples component totals, which cannot express covariance between the categories its stages couple; the NATIONAL run instead samples a parameter graph, and there two physical couplings are carried explicitly—fertiliser N as one audited series driving both the soils components and the urea term, and each species' head-count shared between its enteric and manure components. Their effect at national scale is about a hundredth of a percentage point, in the widening direction. Supplementary Note S3 gives that treatment; the limitation of the sector-scoped stream is the one stated here. The reported national interval therefore reflects the documented parameter uncertainties under this correlation structure. Factor-level uncertainties are reported in Table 3, category distributions in Appendix E, and full input provenance in Supplementary Note S3.

4.6. Key categories, completeness and quality control

Key categories are identified using the IPCC Approach 1 level and trend assessments.

The level assessment ranks categories by their contribution to the absolute national total in the reference year. The trend assessment uses $T_x = L_{x,0}\,|g_x - g_{\mathrm{tot}}|$, where $L_{x,0}$ is category x's share in the base year, g_x is its growth since 1990 and g_{tot} is the growth of the total agricultural inventory. The metric therefore identifies categories that are both important in the base year and whose trajectory differs from the sector average. It should not be interpreted simply as a category growth rate.

Categories are accumulated until the 95 % threshold is reached. A category is classified as key if it qualifies under either the level or the trend assessment.

Inventory coverage is documented through a completeness table. Every AFOLU category-and-gas combination receives an IPCC notation key: estimated, not estimated, not occurring, or included elsewhere. This makes known gaps, exclusions and cross-sector transfers explicit.

Quality control is implemented through automated tests covering units, source traceability, mass-balance reconciliation, benchmark ranges, and consistency among national, provincial, grid, district and historical outputs. Quality assurance is supported through comparison with FAOSTAT, EDGAR, Climate TRACE and independent district statistics. Changes to activity data, emission factors or methods trigger recalculation of the complete time series.

5. Emission Factors and Calculated Emissions

The spatial allocation described in Section 4 determines where emissions are placed but does not change their national magnitude. National totals are determined only by the activity data and the emission factors.

Table 3 reports the principal emission factors, their uncertainty, and their IPCC or literature source. Table 4 reports the country-specific and inventory-specific parameters that replace or refine selected Tier 1 defaults. Full source details and derivations are provided in Supplementary Note S2.

Each factor is stored in a version-controlled data file together with its units, source and uncertainty. This structure allows every calculated value to be traced to a documented input.

5.1. Emission factors

Enteric-fermentation and manure-management factors are drawn mainly from the 2006 IPCC defaults for the Indian Subcontinent. Managed-soils factors use the climate-specific values in the 2019 Refinement.

The direct soil N_2O factor is the most consequential individual methodological choice. The main inventory uses the dry-climate value $EF_1 = 0.005$. The older 2006 aggregate value of 0.010 and the 2019 wet-climate value of 0.016 are retained as sensitivity cases.

The dry-climate value is also close to the crop-weighted empirical estimate of approximately 0.0055 reported by Cui et al.

(2021) for Pakistan. That empirical value is used as a sensitivity anchor, not as the operative emission factor.

Rice methane uses the IPCC baseline factor together with water-regime scaling. The main case assumes continuously flooded conditions, $SF_w = 1$. Intermittent irrigation and rainfed conditions are evaluated as alternatives.

Factor (unit)	Value	±%	Source
Enteric CH_4 (kg/head/yr; 2006 T10.10–10.11)			
Buffalo	55	40	2006 T10.10
Dairy cattle	58	50	2006 T10.11
Other cattle	27	50	2006 T10.11
Camels	46	50	2006 T10.10
Sheep, goats	5	50	2006 T10.10
Manure management (2006 T10.14–10.21)			
CH_4 buffalo, warm (kg/hd/yr)	5	30	2006 T10.14
CH_4 dairy cattle, warm	6	30	2006 T10.14
N_2O EF_3 solid/liquid (kg N/kg N)	0.005	100	2006 T10.21
N_2O EF_3 drylot	0.02	100	2006 T10.21
Rice CH_4 (2006 T5.11–5.13)			
Baseline EF_c (kg/ha/day)	1.30	53	2006 T5.11
SF_w continuous (default)	1.00	26	2006 T5.12
SF_w intermittent (AWD)	0.52	30	2006 T5.12
SF_w rainfed	0.28	40	2006 T5.12
Managed soils (kg N_2O-N/kg N; 2019 Refinement)			
EF_1 dry (default)	0.005	110	2019 T11.1
EF_1 flooded rice	0.004	120	2019 T11.1
EF_1 aggregate (sens.)	0.010	90	2006 default
EF_1 empirical anchor	0.0055	100	Cui et al. (2021)
EF_4 volatilisation (dry)	0.005	100	2019 T11.3
EF_5 leaching	0.011	100	2019 T11.3
Carbon amendments, burning			
Urea carbon (t C/t urea)	0.20	10	2006 Eq.11.13
Residue burning CH_4 (g/kg d.m.)	2.7	40	2006 T2.5

Table 3: Principal emission and carbon factors, with factor-level uncertainty (±%, 95% relative half-width) and provenance. These factor uncertainties are combined with activity-data uncertainty into the per-category values propagated in the uncertainty analysis (Section 4, Appendix E). Values marked *(sens.)* are used only in the sensitivity analysis (Section 5.4); the empirical anchor adjudicates the soil-nitrogen choice and is not the operative factor.

5.2. Country-specific and inventory-specific parameters

The largest departures from global Tier 1 defaults occur in the livestock characterisation.

Population-average typical animal masses are used for dairy cattle, other cattle, buffalo and camels—declared judgements resting on working pointers rather than cited measurements, as Supplement S2's provenance classes record (10 cited, 3 derived, 25 working pointers and 21 declared judgements across the 59 parameter rows). These values are generally higher than the corresponding IPCC defaults and affect both nitrogen excretion and the division of enteric emissions among livestock groups. The dairy-cattle share is derived from FAOSTAT counts of animals reported as being in milk.

Other national parameters determine how fertiliser and soil nitrogen are represented. These include the proportion of synthetic nitrogen supplied as urea, the share applied to rice, and the irrigated fraction used in the leaching calculation.

Table 4 distinguishes among four types of parameter: national activity data, Tier 2 characterisation, regional IPCC defaults, and inventory assumptions. This category-by-category classification defines the methodological basis more accurately than assigning one tier to the whole inventory.

5.3. Calculated emissions

Applying the factors in Table 3 and the parameters in Table 4 to the national activity data produces the 2024 estimates reported in Section 8.

The same emission-factor set is held constant throughout the 1990–2024 reconstruction. Changes in the historical series therefore reflect changes in activity, including livestock growth and increasing fertiliser use, rather than changes in the selected factors.

Appendix C presents the activity data, implied factors and emissions by gas for the 1990 base year and the 2024 reporting year in a Common Reporting Table structure.

5.4. Consequential applicability and structural cases

Non-rice direct N_2O retains the dry-climate factor as a declared reference. Irrigation, rice flooding and grazing conditions require different evidence; national aridity or agreement with FAOSTAT alone does not validate one value for every pathway. Aggregate and climate-factor alternatives remain explicit structural cases without assigned probabilities.

The TAN-coherent reference and IPCC volatilisation alternative each regenerate their nitrogen account. Manure-management shares, excretion rates, residue fractions and dung destination are distinct structural assumptions, while bounded applicable fate fractions are sampled conditionally. Source factors retain their units and native nitrogen-to-gas conversions. Current quantitative intervals and source-category totals are generated from the integrated release rather than the superseded independent nitrogen chains.

6. Shared Nitrogen Accounting and Air-Pollutant Methods

The greenhouse-gas and air-pollutant calculations use a common nitrogen ledger. This ledger tracks how excreted and applied nitrogen moves through manure management, grazing, fertiliser application, volatilisation and soil processes.

The two inventories therefore represent different emissions from the same underlying nitrogen flows. Aligning their system boundaries is necessary to prevent the same nitrogen from being removed twice, retained twice, or assigned inconsistently between NH_3 and N_2O.

6.1. Alignment of the nitrogen boundaries

The reference TAN formulation follows species- and system-specific ammonia loss through housing or yard exposure, storage and field application. Each stage acts on surviving total ammoniacal nitrogen. Management NH_3 and NO_x, leaching, N_2 and N_2O losses are removed once before the surviving organic pool reaches soil. Grazing and field

Parameter	Value (unit)	Status	Source
Dairy (in-milk) cattle share	0.295	held herd-composition split	FAOSTAT QCL 2024 (16.98/57.54 M)
Typical mass, dairy cattle	400 kg	Tier-2 characterisation	Sahiwal/crossbred; LRRD 2006
Typical mass, other cattle	225 kg	Tier-2 characterisation	Pak. adult zebu population avg.
Typical mass, buffalo	380 kg	Tier-2 characterisation	Nili-Ravi herd avg. (LRRD)
Typical mass, camels	480 kg	Tier-2 characterisation	dromedary herd avg.; Khan 2025
Nex, buffalo	0.32 kg N/1000 kg/d	regional IPCC default	IPCC 2006 T10.19 Asia
Nex, dairy cattle	0.47 kg N/1000 kg/d	regional IPCC default	IPCC 2006 T10.19 Asia
Warm-zone share (manure)	0.70	inventory assumption	Indus-plain climate zone
Irrigated applicability fraction	0.80	conditional applicability judgement	approximate irrigation evidence; definitions unresolved
Rice share of synthetic N	0.16	held allocation assumption	crop-area approximation
Urea share of applied N	0.80	early-year fallback	own-year product offtake used where supported
Rice cultivation period	125 days	Tier-2 characterisation	PARC calendar (area-weighted)

Table 4: Country-specific and inventory-specific parameters that override IPCC defaults. *Status* distinguishes national activity data, genuine Tier-2 characterisation, regional IPCC defaults, and inventory assumptions — so the paper does not over-claim a uniform Tier-2 basis. Full provenance (source, table/page, derivation, and uncertainty) for these and all other parameters and factors — management-system shares, dung-cake fraction, residue ratios, water/amendment factors, volatilisation and leaching fractions, and the soil N_2O factor set — is given in Supplementary Note S2.

volatilisation remain separate pathways. The same realised pools drive indirect N_2O and the air species.

A separately identified IPCC alternative uses the applicable management and field volatilisation fractions. It regenerates both product families rather than combining an IPCC N_2O account with a different EMEP ammonia chain. Its management fractions are not additional removal from the TAN reference. Material-flow closure is tested in each configuration and uncertainty draw.

6.2. *Synthetic-fertiliser ammonia*

Synthetic-fertiliser NH_3 is calculated from Pakistan's reported fertiliser-product mix rather than from one generic fertiliser composition.

EMEP/EEA factors vary substantially among products because urea and ammonium-based fertilisers have greater volatilisation potential than nitrate-dominant products, particularly on high-pH soils. Pakistan's alkaline cropland and urea-heavy fertiliser mix make product weighting important.

Let w_j be product j's share of applied nitrogen and $EF_j^{NH_3\text{-}N}$ its ammonia-nitrogen emission factor (EMEP/EEA, 2023b). Taking the NFDC offtake by product (National Fertilizer Development Centre, 2024) over one fiscal year, the product-weighted factor is

$$f_{NH_3,SF} = \sum_j w_j \, EF_j^{NH_3\text{-}N} = 0.1608. \quad (5)$$

Urea supplies approximately 81 % of the nitrogen in the product mix and therefore dominates the weighted result (Table 5). The remaining products change the factor by less than 0.01.

The value 0.1608 is already expressed as kilograms of NH_3-N per kilogram of nitrogen applied. It is converted to NH_3 mass by the ratio 17/14 and used directly in the ammonia calculation. It is not multiplied by an additional NH_3 share.

Soil NO_x is calculated separately, using $f_{NO,SF} = 0.007$ kg NO-N per kilogram of nitrogen applied. For the IPCC indirect-N_2O pathway, which Eq. 11.9 defines on the combined volatilised pool, the two are summed:

$$\mathrm{Frac}_{\mathrm{GASF}} = f_{NH_3,SF} + f_{NO,SF} = 0.1678, \quad (6)$$

against the IPCC "urea-dominant" default of 0.15. Keeping these three quantities separate preserves the nitrogen balance and avoids applying an additional ammonia fraction to a factor already expressed as NH_3-N. Because $f_{NH_3,SF}$ also drives the greenhouse-gas layer's synthetic indirect N_2O, it is resolved in one place and both layers necessarily use it.

6.3. *Livestock ammonia*

Livestock NH_3 is calculated using the EMEP/EEA total-ammoniacal-nitrogen (TAN) mass-flow method (EMEP/EEA, 2023a; Sommer et al., 2019). The IPCC post-application fraction $\mathrm{Frac}_{\mathrm{GASM}}$ is not used as a livestock ammonia emission factor, because it was developed for indirect-N_2O accounting rather than stage-resolved NH_3 estimation.

Excreta deposited during grazing lose a species-specific proportion of TAN directly. Manure entering a managed system can lose ammonia during housing or yard exposure, storage and field application. Each stage acts on the TAN remaining after the preceding stage.

The calculation distinguishes among manure systems. Solid storage passes through housing, storage and field-application stages. Daily-spread manure reaches the field without a storage stage. Dry lots are represented as uncovered yards.

Factors also differ among livestock categories, and are taken from Table 3-9 of the Guidebook. Buffalo manure is assigned a TAN fraction of 0.50, compared with 0.60 for cattle, and a storage factor of 0.17 against 0.32. Grazing factors are applied to TAN rather than to total excreted nitrogen (0.14 for cattle and buffalo, 0.09 for small ruminants).

EMEP/EEA does not provide a separate camel category. Camels therefore use the small-ruminant factors as the closest available published analogue, with a wider uncertainty assigned to reflect the approximation.

6.4. *Other agricultural air pollutants*

Soil NO_x is estimated as approximately 0.7 % of applied synthetic, organic and crop-residue nitrogen, following Stehfest and Bouwman (2006). The value is initially calculated as NO-N and converted to reported NO_x mass using the conventional NO_2-equivalent conversion.

Product	t N	N content	Share of N	$EF^{NH_3\text{-}N}$	Contribution
Urea	3 026 265	0.46	0.8078	0.170	0.1373
DAP	290 777	0.18	0.0776	0.154	0.0120
CAN	206 058	0.26	0.0550	0.043	0.0024
Other NP	202 590	0.20	0.0541	0.154	0.0083
Ammonium sulphate	14 595	0.21	0.0039	0.154	0.0006
NPK	6 162	0.15	0.0016	0.154	0.0003
Total	**3 746 448**		**1.0000**		**0.1608**

Table 5: Derivation of the product-weighted synthetic-fertiliser ammonia factor. The table is generated from `outputs/agriculture_fertiliser_composition_2024.csv`, whose window is exactly the twelve months July 2023 to June 2024; the annexe supplies the normalised composition only, and the nitrogen magnitude remains the audited national offtake. NFDC's national and provincial tables do not sum to each other over the same twelve months: the difference is 15 918 t of product, 0.154 917 % of the national total and mostly urea. It is published by month and product in `outputs/agriculture_nfdc_province_residual.csv` and assigned to Other/unallocated rather than distributed across the provinces, which would make every residual zero and the shares look exact. Product tonnages are NFDC offtake over one fiscal year (National Fertilizer Development Centre, 2024); nitrogen contents and emission factors are EMEP/EEA values for high-pH soils, appropriate to Pakistan's alkaline cropland (EMEP/EEA, 2023b). *All emission factors in this table are kg NH_3-N per kg N applied.* Urea carries 81 % of the nitrogen and therefore dominates the weighting; the audited national nitrogen total (3.775 Mt N) rather than this window's sum is used as the activity datum, with the mix supplying only the composition.

Pollutants from crop-residue burning are calculated from the same dry-matter-burned estimate used in the greenhouse-gas inventory. The burned mass is multiplied by the combustion factor and by pollutant-specific emission factors for CO, $PM_{2.5}$, NMVOC, NO_x, NH_3, black carbon and SO_2 (Table 6). The $PM_{2.5}$ factor is also used by the related satellite-nowcasting layer so that the two products report a consistent crop-burning $PM_{2.5}$ quantity.

All air factors retain native units and source applicability. Local product-rule widths are diagnostics; the integrated conditional ensemble replays shared herd, crop, nutrient and bounded fate-factor realisations across NH_3, NO_x and N_2O. Structural management shares and factor alternatives are separate cases, not confidence limits.

7. Evaluation Framework

The inventory is evaluated through several complementary comparisons. Each addresses a different component of the framework.

First, national and category-level totals are compared with FAOSTAT, EDGAR and Climate TRACE using matched years, gas metrics and category scope. These datasets provide external reference estimates but are not direct observations of national emissions. A ratio between 0.5 and 2.0 is used as a diagnostic screening range. Values outside this range are examined to identify differences in factors, activity data or category definitions. The range is an operational review threshold rather than a formal statistical acceptance criterion.

Second, the model is run for 1990–2023 using FAOSTAT activity series and compared with FAOSTAT emissions. Because the two calculations share much of their activity basis, this comparison primarily tests the implementation and the consequences of different emission factors. It is not an independent test of the underlying activity data.

Third, district spatial patterns are compared with independent government crop statistics that are not used to produce the relevant district allocation. This comparison evaluates the ranking of districts rather than their absolute emissions.

Pollutant / stage	Basis	Factor	±%
NH_3, synthetic fert.	$f_{NH_3,SF}$, kg NH_3-N per kg N	0.1608[b]	30
NH_3, grazing	of TAN; cattle/buffalo | small rum.	0.14 | 0.09	50
NH_3, housing	of TAN; cattle | buffalo | small rum.	0.08 | 0.20 | 0.22	40
NH_3, yard (dry lot)	of TAN; dairy | other cattle	0.30 | 0.53	50
NH_3, storage	of TAN surviving; cattle | buffalo	0.32 | 0.17	40
NH_3, field application	of TAN surviving; cattle | buffalo	0.68 | 0.55	40
TAN fraction	of N excreted; cattle | buffalo | poultry	0.60 | 0.50 | 0.70	20
Soil NO_x	$f_{NO,SF}$, kg NO-N per kg N applied[a]	0.007	100
Crop-residue burning (g/kg dry matter; on $M_{burnt}\cdot C_f$)			
CO	IPCC 2006 T2.5	92	40
$PM_{2.5}$	Akagi et al. (2011)	7.6	50
NMVOC	Akagi et al. (2011) / EMEP	7.0	60
NO_x	IPCC 2006 T2.5	2.5	50
NH_3	Akagi et al. (2011)	1.3	60
BC	Akagi et al. (2011)	0.5	60
SO_2	EMEP/EEA (2019)	0.40	70

Table 6: Air-pollutant factors, calculation basis, and 95% relative uncertainty. Livestock NH_3 follows the species-resolved EMEP/EEA TAN mass-flow method. Crop-burning factors are applied to the same dry matter used in the greenhouse-gas calculation. The synthetic-fertiliser NH_3 factor is already expressed as NH_3-N and receives no additional ammonia-share adjustment. Soil NO_x is calculated as NO-N and reported as NO_2-equivalent mass.

Fourth, reconciliation is tested automatically at every spatial scale. Each grid and district product must reproduce its provincial and national control totals within numerical tolerance. This verifies mass conservation but does not establish the accuracy of the within-province spatial pattern.

Finally, the analytical and Monte-Carlo uncertainty calculations are compared as alternative propagation methods. Since they use the same input assumptions, agreement between them indicates computational consistency rather than independent confirmation.

7.1. District comparison with independent crop records

District allocations are evaluated against district crop-area records published by the Ministry of National Food Security and Research (Ministry of National Food Security and Research, 2023).

Spearman rank correlation is used because the comparison concerns the ordering of districts rather than a claim that crop area directly measures emissions.

The rice-emission ranking is compared with official district rice area. Across 66 matched districts the correlation is $\rho = 0.92$ ($p < 0.001$), and 13 of the 15 highest-ranked rice districts in the inventory are among the 15 highest-ranked districts in the official data.

Managed-soils emissions are compared with district wheat area across 85 matched districts. Wheat is used as a broad indicator of cultivated and fertilised land because it is grown across most cropping districts. The resulting rank correlation is $\rho = 0.84$.

Agreement varies geographically. It is strongest in Punjab and Balochistan and weak in Sindh, where managed-soils emissions follow a rice-and-cotton geography that is not well represented by wheat area alone. The Sindh result therefore reflects a limitation of the comparison variable as well as possible uncertainty in the allocation.

Equivalent independent district checks are not possible for all categories. Livestock emissions already use PBS district census counts where available, so comparison with the same counts would not be independent. No comparable published district dataset was identified for field burning or urea application. These categories are consequently recorded as lacking independent district evaluation rather than being assigned a validation score.

Category	Independent record	n	Spearman ρ
Rice cultivation	District rice area (MNFSR)	66	0.92
Managed soils	District wheat area (MNFSR)	85	0.84
Enteric / manure	— (census is the proxy)	—	n/a
Field burning	— (no district record)	—	n/a
Urea application	— (no district record)	—	n/a

Table 7: Rank comparison between modelled district emissions and independent government crop-area records. Rice emissions are compared with district rice area, and managed-soils emissions with district wheat area. Categories for which the only available district statistic is already used as an input, or for which no suitable district record exists, are reported as not independently evaluated. Generated by `src/pkinv/eval/validation/district_rice_records.py` and `district_records_multi.py`; the same comparison is scored as a registered benchmark task in `outputs/benchmark_spatial_allocation.csv`.

7.2. What each comparison shares with the engine

External is not the same as independent, and a comparison that tuned a product cannot also test it. Table 8 states, for every comparison this section and the release's validation catalogue carry for agriculture, what the comparison shares with the inventory and whether it is a calibration. Three rows carry a weaker claim than they did before this release, and in each case the reason is that the comparison reads something the product is built from.

8. Results

The TAN-coherent agricultural reference produces 223 Mt CO_2-equivalent on AR5 GWP-100 in FY2023–24 and 228.6 Mt in FY2024–25. Native CO_2, CH_4 and N_2O remain separately reported. Crop and nutrient activity uses compatible fiscal controls; livestock is a consistent reconstruction around the adopted census rather than an annually observed herd.

8.1. Spatial products

Livestock, cropland, nutrient, rice and fire supports allocate different processes. Administrative controls and placed/unallocated mass retain their own boundaries. A fixed source surface changes magnitude with activity but does not establish changing local activity. The grid's nominal 0.05° spacing describes the allocation product, not an observational resolution. Independent district crop records test selected crop patterns; livestock counts used to construct a proxy cannot validate that same proxy.

8.2. Temporal change and uncertainty

The common series covers ten fiscal years, July 2015–June 2025. Legacy 1990–2024 calendar diagnostics remain separately labelled and are not spliced onto the new census reference to make a common national series. Trends use paired realisations and shared source revisions. Conditional uncertainty reflects sampled inputs and fate factors; management shares, factor applicability, material destinations and geographic/timing uncertainty are structural or unestimated limitations.

8.3. Nitrogen and residue accounting

TAN-reference stage losses determine surviving manure pools, management and grazing NH_3/NO_x and the corresponding indirect N_2O. The IPCC alternative regenerates the same products on its separate parameter formulation. The two configurations are not combined into one probabilistic interval. Residue supply is partitioned among returned, field-burned, household-fuel and unresolved destinations. Closing those accounts does not observe their split. Household dung reference S receives the survey-supported quantity; alternative L assigns all diverted material to household combustion.

The air results in Section 9 show why retaining the nitrogen pathways matters: the common ledger includes NO_x removed during management and grazing alongside soil and burning sources, while distinct product factors drive fertiliser NH_3. External comparisons assess matched source boundaries and identify shared inputs rather than selecting a factor to match an external total.

Comparison	What it tests	Shared with the engine	Independence	Calibration
FAOSTAT / EDGAR / CRT categories	Method and emission factors	The national activity statistics both sides read	Partial	No
Cui 2021 direct-N_2O	Direction of the EF1 choice	None stated; the Pakistani value is extrapolated with no in-country observation	Partial	No
District rice area (MNFSR)	Spatial rank of rice emissions	Nothing: the allocation does not read crop area	Large	No
District wheat area (MNFSR)	Spatial rank of managed soils	Nothing	Large	No
Livestock district census	—	The census counts ARE the activity	None	No
NFDC monthly urea offtake	Spread between two monthly shapes	The offtake series IS the adopted profile	None	Yes
VIIRS/MODIS fire detections	Attribution of detections to Pakistani cropland	The detections ARE the burning profile's weights	None	Yes
FAO rice crop calendar	Plausibility of the rice window	Calendar conventions	Partial	No

Table 8: Independence and shared-input matrix for the agriculture comparisons, in the form the aviation paper uses. A comparison whose "shared" column names the product's own input can bound allocation or attribution, and cannot confirm the quantity: the monthly fertiliser and fire rows are calibrations for that reason, and the enteric and manure categories are recorded above as lacking an independent district record rather than being scored against the census that supplies them.

Source category (Mt CO_2-equivalent)	FY2023–24	FY2024–25
Enteric fermentation	151.7	156.4
Field burning of residues	0.6705	0.6182
Managed soils	28.49	27.78
Manure management	20.84	21.66
Rice cultivation	16.55	17.74
Urea application	4.824	4.319

Table 9: Current agricultural source-category totals under S_TAN, generated from the native annual records.

9. Agricultural air-pollutant results and evaluation

Agriculture contributes 2619 kt NH_3 and 318 kt NO_2-equivalent NO_x in FY2023–24 under S_TAN. The agricultural values below are covered-source totals, not proof that every agricultural emission process is estimated. Black carbon is a component of $PM_{2.5}$ and must not be added to it.

Species (Gg)	FY2023–24	FY2024–25
$PM_{2.5}$	54.13	49.9
BC	3.561	3.283
NO_x	318	318.4
SO_2	2.849	2.626
CO	655.2	604.1
NMVOC	49.85	45.96
NH_3	2619	2639

Table 10: Current agricultural air quantities from the common fiscal release under the TAN-coherent reference. Values by process and pollutant are in `outputs/air_pollutants_2024.csv`.

Housing, storage and yards contribute about 838 Gg NH_3, applied manure 815 Gg, synthetic fertiliser 737 Gg and pasture deposition 218 Gg in the reference year. Crop burning adds about 9.3 Gg. These stages act on the same conserved pools as agricultural N_2O. The air NO_x total includes management and grazing losses as well as synthetic, organic and crop-residue soil sources and burning; excluding management/grazing NO_x while retaining their volatilised nitrogen in the N_2O calculation would describe different ledgers.

9.1. Temporal and spatial support

Livestock housing/storage follows livestock support; applied manure follows the livestock–cropland intersection; synthetic fertiliser follows nutrient/cropland support. Field burning and household residue combustion use separate locations and activities. The 0.05° cells are allocations on those supports, not observations at that resolution. Placed plus unallocated mass closes to each pollutant's compatible control.

Manure-stage, grazing and residue-decomposition emissions have no observed monthly release series and use declared days-weighted allocation. Fertiliser offtake supplies a timing proxy, not observed application or volatilisation. Crop-burning fire evidence is a seasonal/location proxy and does not measure the national burned fraction. Temporal and spatial evidence shares are pollutant-specific. Newly extended annual activity does not imply newly observed monthly geography.

9.2. Conditional and structural uncertainty

The integrated conditional ensemble uses the same realised herd, crop production, nutrient offtake and applicable nitrogen-fate factors for NH_3, NO_x and N_2O. Bounded fractions prevent negative remaining material or emission fractions above one. Trend intervals use paired years, with persistent and annually independent activity-error cases kept separate. The intervals do not include alternative manure-management shares, excretion rates, climate-factor choices, missing sources or allocation uncertainty.

The TAN reference and IPCC alternative each regenerate their nitrogen account. Regional urea-factor and manure-management sensitivities remain structural cases without assigned probabilities. Agreement with FAOSTAT or another inventory is not a selection rule for a soil factor. The dry-climate non-rice factor is a declared reference whose applicability to irrigated cropland is imperfectly established; rice and grazing retain their own formulations.

9.3. Coverage

Source process	Supported air products
Housing, yards and storage	NH_3 and management NO_x
Grazing and manure application	NH_3 and source-specific NO_x
Synthetic fertiliser and crop-residue soil inputs	NH_3/NO_x; residue NH_3 is a calculated threshold result
Field residue burning	$PM_{2.5}$, BC, NO_x, SO_2, CO, NMVOC, NH_3
Agricultural machinery; tillage/harvest dust	Air estimates not admitted; explicit missing-source cells

Table 11: Estimated processes and material air gaps. An estimated module cell can coexist with an unestimated process.

NH_3 from residues left in fields is calculated with the source's nitrogen-content threshold equation. A zero under the reference contents is a model result, not observed absence; nitrogen-content alternatives test that threshold. Coarse particulate and additional agricultural sources are not invented to complete a species rectangle.

9.4. Matched external evidence

The original public HTAPv3.2 NH_3 country workbook provides annual and monthly source controls without a private account. Pakistan's agriculture combines livestock, crops and agricultural waste burning. Exact July–June sums support comparisons through FY2019–20; later fiscal years are blank because the released HTAP series ends in calendar 2020. For FY2019–20, S_TAN agriculture is 2215 Gg against 2184 Gg in HTAP, a ratio of 1.014. This is a sector-boundary comparison, not independent verification of individual pathways.

The HTAP mosaic uses REAS for much of South Asia and EDGAR for gap filling, agricultural burning and post-2015 trends. Shared inputs and trend extensions limit independence. Differences are interpreted by activity, livestock fate, product-specific fertiliser factors and coverage. No parameter is adjusted solely to reproduce the external total. Spatial coincidence with a proxy used in both products is not independent validation, and satellite ammonia columns require transport and retrieval treatment before they can constrain emissions.

10. Discussion

Pakistan-specific activity and process boundaries determine the inventory's contribution. Crop, nutrient and census-anchored herd controls support a national reconstruction whose source pathways and geographic support remain inspectable. The common nitrogen ledger permits differences in NH_3, NO_x and N_2O to be explained by pathway or factor rather than hidden in incompatible accounts.

The retained dry-climate soil factor is consequential and conditional. Irrigated cropland, flooded rice and grazing have different factor applicability, so a national climate label does not resolve every pathway. Aggregate and climate-factor sensitivities show the consequence without fitting to FAOSTAT or national reporting. The same distinction applies to manure-system shares: material closure checks the calculation but does not establish the chosen management distribution.

Independent district crop records evaluate selected spatial patterns. Inventories constructed from the same livestock counts or cropland surfaces do not independently confirm those allocations. Coarse support downscaled to 0.05° retains proxy limitations. Source-matched public HTAP comparisons supplement national reporting and global inventories, with provider and shared-input provenance explicit.

The common core's latest annual year and its detailed reference year have different roles. FY2024–25 uses supportable annual activity; it does not acquire new monthly application dates, herd enumeration or fine-scale source geography automatically. Fitness for use therefore depends on species, emitting process, year and configuration as well as total mass. Atmospheric concentrations, exposure and health effects require additional evidence and are outside this inventory.

11. Limitations

Livestock levels are reconstructed consistently around the adopted census, but annual fiscal head counts are not annual enumerations. Manure-system shares, nitrogen excretion and the destination of diverted dung remain structurally uncertain. Reference S receives the observed household fuel quantity; unresolved diverted material remains outside an assigned combustion fate. L assumes all diverted material is burned in households. A closed transfer does not establish the destination.

Tier-1/default factors and selected country-specific parameters do not constitute Pakistani field measurements. The direct non-rice soil factor's dry-climate transfer is consequential in irrigated cropland; rice and grazing require separate assessment. The aggregate soil-factor alternative is published without tuning to FAOSTAT. Continuous flooding is a rice-water-regime assumption rather than an annual management observation.

Fertiliser offtake is distinct from application. Product composition and nutrient controls are reconciled, but application dates, soil conditions and manure placement remain imperfectly observed. Regional urea factors and source-stage TAN factors may transfer differently under Pakistan's conditions. A common nitrogen ledger prevents incompatible pathways; it does not validate those factors.

Crop production constrains residue supply. Burned, returned, household-fuel and unresolved residue destinations are distinct. Burning shares, combustion completeness, nitrogen contents and emissions factors remain uncertain; fire detections do not directly measure mass burned. Agricultural machinery and tillage/harvest dust lack admitted air estimates.

Conditional uncertainty samples shared inputs and bounded fate parameters. Fixed management shares, structural factor choices, unestimated categories and geographic/timing uncertainty are excluded from those percentiles and shown separately. Reference-year spatial patterns and monthly

proxies are labelled as allocations. National source magnitudes cannot be interpreted as ambient concentration or exposure rankings. The inventory remains non-LULUCF; exploratory land-sector material is outside the joint national totals.

12. Conclusions

The maintained agricultural inventory reconstructs Pakistan-specific crop, nutrient and census-anchored livestock activity within the common July 2015–June 2025 core. GHG and air products share their nitrogen and residue flows while preserving different emission equations, chemical mass conventions and geographic supports.

The reference TAN configuration yields 223 Mt CO_2-equivalent, 2619 kt NH_3 and 318 kt NO_2-equivalent NO_x in FY2023–24. Management and grazing NO_x are included consistently with the volatilised nitrogen used for indirect N_2O. The IPCC configuration, regional factor and management cases remain separate structural products.

Shared conditional realisations preserve nitrogen closure and paired historical changes. Missing machinery/dust sources, climate-factor applicability, manure shares, crop-residue fates and allocation uncertainty remain explicit. Direct public HTAP source tables permit traceable matched-year comparisons, while shared inputs limit independence. These products provide source-specific activity and pathway information beyond a clipped global inventory without claiming field validation of every factor or an exposure assessment.

Code and data availability

The full system — the calculation engine for both the greenhouse-gas and air-pollutant layers, the shared nitrogen ledger, the gridding and district pipelines, the versioned data files, the test suite and the outputs reported here — is version-controlled in the inventory repository (https://github.com/PakAirQuality/pakistan-emissions-sectors).
It reproduces from a single command (`run_all.py`) with NO user-supplied input: the FIRMS extraction is committed and hash-pinned, and the MapSPAM rice raster is held and hash-pinned beside it (data/gis/manual/RICE_SOURCES.json) with its cropland-fallback class deleted from the allow-list, so a checkout without the raster refuses the build rather than silently substituting (Appendix A). DOIs are minted at archival and none exists yet. Activity data derive from public sources listed in Section 3; downloaded caches are regenerated by the pipeline. The independent ammonia grid used for spatial validation is redistributed by its authors under an open licence (Tomlinson et al., 2025).

References

Ahmad, R., 2026. PakEmis 1.0.0rc69: greenhouse gas and air pollutant emissions in pakistan, july 2015–june 2025. National data-description manuscript with technical report and sector notes; local review package prepared. External repository review deposit and persistent identifier pending; no journal submission asserted.

Akagi, S.K., Yokelson, R.J., Wiedinmyer, C., Alvarado, M.J., Reid, J.S., Karl, T., Crounse, J.D., Wennberg, P.O., 2011. Emission factors for open and domestic biomass burning for use in atmospheric models. Atmospheric Chemistry and Physics 11, 4039–4072. doi:10.5194/acp-11-4039-2011.

Climate TRACE, 2026. Climate TRACE emissions inventory and data downloads. https://climatetrace.org/data. Version 7 country- and source-level API records; CC BY 4.0; accessed 6 August 2026.

Crippa, M., Guizzardi, D., et al., 2024. GHG emissions of all world countries — 2024 report (EDGAR GHG booklet, prior vintage). European Commission, Joint Research Centre. Prior-vintage booklet, retained only for the agriculture CY2023 comparison row (`data/benchmarks.csv` row 8); source file not held, no hash pinned. Superseded for every other comparison by *edgar2025booklet*. https://edgar.jrc.ec.europa.eu/.

Crippa, M., Guizzardi, D., et al., 2025. GHG emissions of all world countries — 2025 report (EDGAR GHG booklet). European Commission, Joint Research Centre. Sheet GHG_by_sector_and_country; annual series 1970–2024; CC BY 4.0. Snapshot archived at `sources/EDGAR/EDGAR_2025_GHG_booklet_2025.xlsx`, sha256 `8bd6b8a4...` pinned per benchmark row in `data/benchmarks.csv`. https://edgar.jrc.ec.europa.eu/.

Crippa, M., Solazzo, E., Huang, G., Guizzardi, D., Koffi, E., Muntean, M., Schieberle, C., Friedrich, R., Janssens-Maenhout, G., 2020. High resolution temporal profiles in the Emissions Database for Global Atmospheric Research. Scientific Data 7, 121. doi:10.1038/s41597-020-0462-2.

Cui, X., Zhou, F., Ciais, P., Davidson, E.A., et al., 2021. Global mapping of crop-specific emission factors highlights hotspots of nitrous oxide mitigation. Nature Food 2, 886–893. doi:10.1038/s43016-021-00384-9.

EMEP/EEA, 2019. EMEP/EEA air pollutant emission inventory guidebook 2019 — 3.F Field burning of agricultural wastes. Technical Report. European Environment Agency.

EMEP/EEA, 2023a. EMEP/EEA air pollutant emission inventory guidebook 2023 — 3.B Manure management. Technical Report. European Environment Agency. Copenhagen. URL: https://www.eea.europa.eu/publications/emep-eea-guidebook-2023.

EMEP/EEA, 2023b. EMEP/EEA air pollutant emission inventory guidebook 2023 — 3.D Crop production and agricultural soils. Technical Report. European Environment Agency. Copenhagen. URL: https://www.eea.europa.eu/publications/emep-eea-guidebook-2023.

FAO, 2025. FAOSTAT: Emissions totals, crops and livestock, fertilizers by nutrient, and forestry production and trade (wood fuel). Food and Agriculture Organization, Rome. https://www.fao.org/faostat/.

Finance Division, Government of Pakistan, 2026. Pakistan Economic Survey 2025–26. Technical Report. Ministry of Finance. Islamabad. URL: https://www.finance.gov.pk/survey_2026.html.

Giglio, L., Schroeder, W., Justice, C.O., 2016. The collection 6 MODIS active fire detection algorithm and fire products. Remote Sensing of Environment 178, 31–41. doi:10.1016/j.rse.2016.02.054.

Gilbert, M., Nicolas, G., Cinardi, G., Van Boeckel, T.P., Vanwambeke, S.O., Wint, G.R.W., Robinson, T.P., 2018. Global distribution data for cattle, buffaloes, horses, sheep, goats, pigs, chickens and ducks in 2010. Scientific Data 5, 180227. doi:10.1038/sdata.2018.227.

Gilbert, M., Nicolas, G., Cinardi, G., et al., 2024. Gridded livestock of the world (glw 4), reference year 2020. Food and Agriculture Organization. https://data.apps.fao.org/catalog/.

Global Change Impact Studies Centre, 2019. Greenhouse Gas Inventory of Pakistan. Technical Report. GCISC, Ministry of Climate Change, Islamabad.

International Food Policy Research Institute, 2024. MapSPAM spatial production allocation model, SPAM 2020 v2r2. Harvard Dataverse. doi:10.7910/DVN/SWPENT.

IPCC, 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use. Technical Report. Institute for Global Environmental Strategies, Japan.

IPCC, 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Technical Report. IPCC, Switzerland.

Ministry of Finance, 2024. Pakistan Economic Survey 2023–24, Chapter 2: Agriculture. Technical Report. Government of Pakistan.

Ministry of National Food Security and Research, 2023. Crops Area and Production (District-wise) 2022–23. Technical Report. Government of Pakistan, Economic Wing. Islamabad.

Mir, K.A., Purohit, P., Cai, S., Kim, S., 2021. Comparative analysis of greenhouse gas emission inventory for pakistan: AFOLU sector. Advances in Climate Change Research 12, 132–144. doi:10.1016/j.accre.2021.01.003.

NASA, 2025. Fire information for resource management system (FIRMS): VIIRS active fire. NASA Earth Science Data Systems. https://firms.modaps.eosdis.nasa.gov/.

National Fertilizer Development Centre, 2024. Fertilizer Review 2023–24. Technical Report NFDC Publication No. 2/2024. Ministry of National Food Security and Research, Government of Pakistan. Islamabad.

National Fertilizer Development Centre, 2025. Fertilizer Review 2024–25. Technical Report. Planning Commission, Government of Pakistan. Islamabad.

Oda, T., Maksyutov, S., Andres, R.J., 2018. The open-source data inventory for anthropogenic CO2, version 2016 (ODIAC2016): a global monthly fossil fuel CO2 gridded emissions data product for tracer transport simulations and surface flux inversions. Earth System Science Data 10, 87–107. doi:10.5194/essd-10-87-2018.

Pakistan Bureau of Statistics, 2025. 7th Agricultural Census 2024. Technical Report. Government of Pakistan.

Runfola, D., Anderson, A., Baier, H., et al., 2020. geoBoundaries: A global database of political administrative boundaries. PLOS ONE 15, e0231866. doi:10.1371/journal.pone.0231866.

Sommer, S.G., Webb, J., Hutchings, N.D., 2019. New emission factors for calculation of ammonia volatilization from european livestock manure management systems. Frontiers in Sustainable Food Systems 3, 101. doi:10.3389/fsufs.2019.00101.

Stehfest, E., Bouwman, L., 2006. N_2o and no emission from agricultural fields and soils under natural vegetation: summarizing available measurement data and modeling of global annual emissions. Nutrient Cycling in Agroecosystems 74, 207–228. doi:10.1007/s10705-006-9000-7.

Tomlinson, S.J., Carnell, E.J., Pearson, C., Sutton, M.A., Jain, N., Dragosits, U., 2025. Gridded emissions of ammonia (NH3) from agricultural sources in South Asia at 0.1 degrees resolution, 2015. doi:10.5285/e0114a4f-32c2-41d9-9c2a-c46f365d4c30.

Tomlinson, S.J., Carnell, E.J., Pearson, C., Sutton, M.A., Jain, N., Dragosits, U., 2026. A framework for gridded estimates of ammonia emissions from agriculture in south asia. Earth System Science Data 18, 2349–2370. doi:10.5194/essd-18-2349-2026. south Asian Nitrogen Hub; country totals in Table 3.

Tubiello, F.N., Salvatore, M., Rossi, S., Ferrara, A., Fitton, N., Smith, P., 2013. The FAOSTAT database of greenhouse gas emissions from agriculture. Environmental Research Letters 8, 015009. doi:10.1088/1748-9326/8/1/015009.

Zanaga, D., Van De Kerchove, R., Daems, D., et al., 2022. ESA WorldCover 10 m 2021 v200. Zenodo doi:10.5281/zenodo.7254221.

Appendix A. Reproducibility

The inventory runs from a documented pipeline (`python3 run_all.py`) that executes the agricultural stages — national inventory, provincial downscaling, gridding, district downscaling, time series, Monte-Carlo, key-category, sensitivity, and uncertainty-input analysis, completeness table, and a consolidated dashboard — and writes all tables and figures reported here. The exploratory land sector is an *optional* module (skipped under `-quick`; see Supplementary Note S1) and feeds none of the agricultural results. The agricultural core runs on the Python standard library alone; the gridding and figures use an optional geospatial stack. The shipped surfaces need nothing supplied: the FIRMS extraction and the MapSPAM rice raster are both committed-or-held and hash-pinned, so "one command" means one command (the default contract is set out below). Every emission factor, parameter, and activity value is stored in a versioned data file, and a suite of more than 240 unit tests checks the IPCC worked-example calculations, exact reconciliation at each scale (national, provincial, gridded, district, and the district × year panel), emission-factor traceability, and the key-category, sensitivity, uncertainty, and completeness outputs. The suite also covers the air-pollutant engine and its shared nitrogen ledger — ledger closure at every stage, the separation of the pre-field and post-application volatilisation fractions, conservation between the NH_3-N and NH_3+NO_x bases, the province and grid allocation of every pollutant, the matched-year and spatial benchmarks, and the structural manure-management scenarios — together with a consistency check that every headline number and figure in this manuscript still matches the output file it was generated from.

Code and data availability.. The full source code, data files and tests are version-controlled; DOIs are minted at archival (none exists yet, and no identifier is printed before it resolves), and the archived release corresponding to any submitted version will let reviewers run the same code that produced these numbers. The archive contains the calculation engine for both layers, the shared nitrogen ledger (`outputs/n_ledger_2024.csv`, every stage-flow row as kilograms of nitrogen and as a share of what was excreted) and its diagnostic runs (`outputs/n_ledger_stage_comparison.csv`, the IPCC and EMEP ammonia-nitrogen factors set against each other stage by stage), the versioned emission-factor and activity files, the gridded (GeoTIFF, EPSG:4326) and district (CSV) products, the twelve-month layer for the reference year (`outputs/agriculture_monthly_2024.csv`, each month's share with the emissions it implies and the basis of the shape) with the timing class each category's evidence supports (`outputs/agriculture_timing_classes.csv`), the crop-residue ledger recording where each tonne of residue goes (`outputs/agriculture_crop_residue_ledger_2024.csv`), the 1990–2024 time series, the uncertainty and benchmark outputs—including the reference-year greenhouse-gas Monte Carlo by category (`outputs/uncertainty_montecarlo.csv`), the level and trend key-category assessment (`outputs/key_categories.csv`) and `outputs/air_uncertainty_montecarlo.csv`, the per-pollutant intervals with the Approach 1 comparison beside them—and the test suite; units, coordinate reference system and time coverage are recorded with each product, and a run manifest records the exact inputs behind the tables reported here. Reproduction: install the (optional) geospatial stack and run `python3 run_all.py`. The shipped surfaces reproduce as-is with no user-supplied input; the contract is stated in full in Appendix B. Expected outputs are the CSV tables and PNG figures reproduced here (written under `outputs/`). The deposited capsule carries its own single command, `sh capsule/run_tests.sh`, which reruns the agriculture suite in a fresh interpreter with nothing installed; in the full repository `python3 -m pytest tests` runs the test suite, as does `python3 -m unittest discover tests`, which is what continuous integration uses. On the released commit both pass in full; on any other commit the release-integrity tests fail by design: they compare the committed release artefacts—manifest, cover, checksums, PDFs—against the working tree, and they can only pass on the exact commit the artefacts were cut from. They are not a failed reproduction, and a reader who sees them should not read them as one. The code is released under the MIT licence and the derived data under CC-BY-4.0; third-party inputs retain their original licences.

Appendix B. Data acquisition

THE REPRODUCTION CONTRACT, stated once. Livestock density (GLW 4, 2020) is retrieved directly from the FAO open cloud store. The FIRMS VIIRS extraction is *committed* and hash-pinned in `data/gis/manual/`, so no access key is needed to reproduce (a key is needed only to RE-FETCH, which would create a new extraction). MapSPAM 2020 is *held* (V2r2, hash-pinned in `RICE_SOURCES.json`) and its former cropland-fallback class is deleted from the allow-list, so a checkout missing the raster refuses the build rather than silently substituting. The DEFAULT reproducible surface is therefore: MapSPAM for rice, frozen FIRMS for burning, and nothing to supply. Re-fetching FIRMS or swapping the rice surface (the own-year GCD-Rice 2023 classification ships as the priced alternative) changes the surface and is an *upgrade*: it creates a new configuration and a new manifest, and its outputs are not the ones reported here. The pipeline auto-detects supplied files and logs which proxy each category used.

Appendix C. Reporting-category and period conventions

The current national core uses fiscal years from July 2015 through June 2025. Its detailed agricultural reference is FY2023–24; the adopted census supplies livestock stocks, while crops and nutrient offtake retain their stated fiscal periods. The former calendar-1990 diagnostic remains a separate historical calculation and is not combined with this core as an observed fiscal series.

IPCC 2006 category	Native gas	Shared source account
3.A.1 Enteric fermentation	CH_4	Census-anchored livestock history
3.A.2 / 3.C.6 Manure management	CH_4, N_2O	Excreted N, management and surviving material
3.C.7 Rice cultivation	CH_4	Own-year rice area and declared water regime
3.C.4 / 3.C.5 Managed soils	N_2O	Synthetic, organic, pasture and crop-residue nitrogen
3.C.1.b Field burning	CH_4, N_2O	Produced residue, burned fraction and completeness
3.C.3 Urea application	CO_2	Product-specific national urea offtake

Table 12: Current agricultural reporting accounts. Native annual records supply quantities and separate AR5/AR6 aggregates under each nitrogen formulation. The household dung-fate choice affects the linked Energy account; source-process coverage and notation statuses remain explicit.

The nitrogen ledger identifies management losses before field application and post-application losses separately. TAN and complete IPCC formulations are named structural choices. Fertiliser offtake is not observed application timing; a calendar crop or herd observation does not become a monthly emission measurement. The residue ledger publishes returned, field-burned, fuel and unresolved material, with dry-to-air-dry moisture conversion and combustion completeness applied at their own boundaries. None of these flows is inferred from a historical total copied into a reporting table.

Appendix D. Completeness and QA/QC

Table 13 gives the notation-key coverage of the agriculture categories. All major agriculture source categories are represented, with the not-estimated (NE) gaps shown explicitly: soil-organic nitrogen mineralisation (F_{SOM}), which requires field data; and liming, set to NE (assumed negligible on Pakistan's alkaline, calcareous soils pending national lime-use data) rather than assumed not to occur. Dung-cake combustion is the one cross-sector transfer to Energy. Land-sector completeness (forest, cropland/grassland SOC, wetlands, harvested wood products) is documented in Supplementary Note S1. Air-pollutant co-emissions (NH_3, NO_x, $PM_{2.5}$, CO, NMVOC, BC) are estimated for the relevant categories in Section 9 and reported in native mass, outside this CO_2e completeness table; agricultural SO_2 is estimated for supported residue burning only. Quality control is automated (source traceability, unit checks, exact reconciliation, benchmark bands); quality assurance is the independent FAOSTAT and EDGAR comparison; and any method or factor change — such as the harmonisation of the volatilisation factor EF_4 to the dry-climate value across the manure and soils terms — triggers a full recalculation of every supported common-core fiscal year to preserve time-series consistency.

Appendix E. Conditional uncertainty inputs

The current joint ensemble is reproduced from the registered activity, factor and conditional-parameter products. Native manure/soil nitrogen is replayed with common herd, crop, nutrient and bounded fate-factor draws. Waste fate and applicable fugitive activity use their physical native reruns. Shared identities refer to actual inputs rather than merely named correlations.

Parameter widths, distribution, mean where required, provenance, seed and draw count are recorded with the release. Paired historical realisations yield trend intervals. Management-system shares, excretion rates, structural soil/climate factors, missing categories and temporal/spatial allocation uncertainty are excluded from the conditional percentiles. The older aggregate category-lognormal diagnostic does not define the current uncertainty product and is retained only with the preserved prior release.

Category	**Gas**	**Key**
3.A Enteric fermentation	CH_4	E
3.B Manure management	CH_4, N_2O	E
3.C Rice cultivation	CH_4	E
3.D Agricultural soils	N_2O	E
SOC mineralisation (F_{SOM})	N_2O	NE
3.E Savanna burning	—	NO
3.F Field burning of residues	CH_4, N_2O	E
3.G Liming	CO_2	NE
3.H Urea application	CO_2	E
Dung-cake combustion	multi	IE

Table 13: Completeness and notation keys for the agriculture sector (IPCC Vol. 1 Ch. 8). E estimated, NE not estimated, NO not occurring, IE included elsewhere (Energy). Land-sector categories are in Supplementary Note S1.